DECODELABS | DATA ANALYTICS | BATCH 2026

Project 3
SQL Data Analysis

Extracting business insights from 1,200 e-commerce orders

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Dataset: Dataset_for_Data_Analytics_3.xlsx

Skills: SELECT, WHERE, ORDER BY, GROUP BY, HAVING, COUNT / SUM / AVG

1. Project Overview

Objective: Use SQL queries to extract insights from an e-commerce orders dataset, applying filtering, grouping, sorting and aggregation.

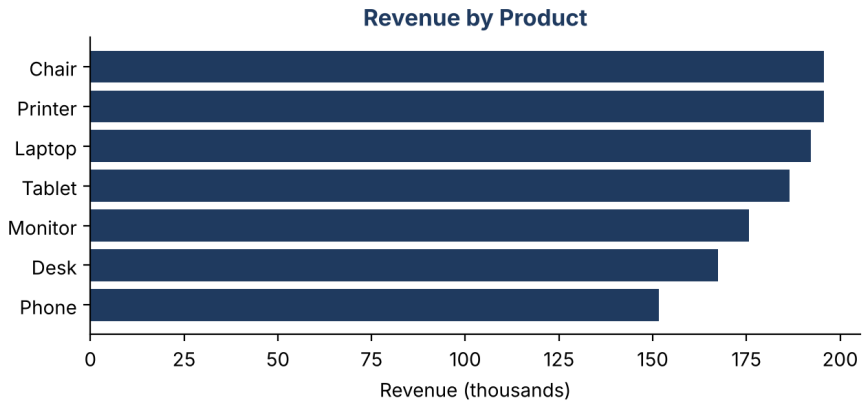
Dataset: 1,200 orders from 2023 to 2025 with 14 columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice. The sheet was loaded as a table named `orders`.

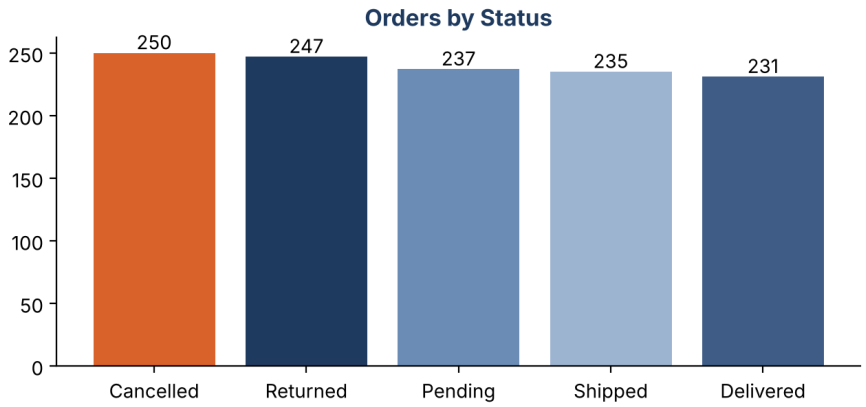
Method: 17 queries were written and executed in SQLite. Each query below shows the SQL, its result and a short business insight.

1,200	1,264,762	1,053.97	3,535
Total Orders	Total Revenue	Avg Order Value	Units Sold

Key Findings

- Only about 39% of orders reached Shipped or Delivered; about 41% were Cancelled or Returned.
- Revenue is evenly spread across seven products; Chair and Printer lead at 15.47% each.
- Instagram is the strongest referral source and Credit Card the top payment method by revenue.
- The data is clean: no duplicate orders and every TotalPrice equals Quantity x UnitPrice.





2. Queries and Results

Q1. Preview the data: first 10 orders

```
SELECT OrderID, Date, Product, Quantity, UnitPrice, OrderStatus, TotalPrice
FROM orders
LIMIT 10;
```

OrderID	Date	Product	Quantity	UnitPrice	OrderStatus	TotalPrice
ORD200000	2023-01-04	Monitor	5	570.62	Shipped	2,853.10
ORD200001	2024-08-23	Phone	2	151.35	Shipped	302.70
ORD200002	2024-02-27	Tablet	5	550.68	Cancelled	2,753.40
ORD200003	2023-10-15	Chair	1	273.19	Returned	273.19
ORD200004	2025-05-08	Printer	4	626.01	Delivered	2,504.04
ORD200005	2023-10-23	Phone	2	245.86	Shipped	491.72
ORD200006	2025-06-17	Laptop	1	664.42	Returned	664.42
ORD200007	2023-05-12	Monitor	5	149.55	Shipped	747.75
ORD200008	2025-04-02	Phone	2	134.28	Cancelled	268.56
ORD200009	2023-11-21	Desk	4	509.38	Shipped	2,037.52

Insight: Each row is one order with product, quantity, unit price, status and total.

Q2. Total number of orders

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

total_orders
1200

Insight: The dataset contains 1,200 orders.

Q3. Overall KPIs: revenue, average order value, units sold

```
SELECT COUNT(*) AS total_orders,
       ROUND(SUM(TotalPrice),2) AS total_revenue,
       ROUND(AVG(TotalPrice),2) AS avg_order_value,
       SUM(Quantity) AS units_sold
FROM orders;
```

total_orders	total_revenue	avg_order_value	units_sold
1200	1,264,761.96	1,053.97	3535

Insight: Total revenue is 1,264,761.96 with an average order value of 1,053.97 across 3,535 units.

Q4. Orders by status (count)

```
SELECT OrderStatus, COUNT(*) AS orders
FROM orders
GROUP BY OrderStatus
ORDER BY orders DESC;
```

OrderStatus	orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Insight: Only 466 orders (about 39%) are Shipped or Delivered; Cancelled and Returned together make up 497 orders (about 41%).

Q5. Revenue and order count by product

```
SELECT Product,
       COUNT(*) AS orders,
       SUM(Quantity) AS units_sold,
       ROUND(SUM(TotalPrice),2) AS revenue,
       ROUND(AVG(TotalPrice),2) AS avg_order_value
FROM orders
GROUP BY Product
ORDER BY revenue DESC;
```

Product	orders	units_sold	revenue	avg_order_value
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58

Insight: Chair and Printer lead on revenue (about 195.6K each); Phone is the weakest product.

Q6. Revenue by payment method

```
SELECT PaymentMethod,
       COUNT(*) AS orders,
       ROUND(SUM(TotalPrice),2) AS revenue
FROM orders
GROUP BY PaymentMethod
ORDER BY revenue DESC;
```

PaymentMethod	orders	revenue
Credit Card	234	263,847.63
Online	258	262,442.94
Cash	246	259,786.29
Gift Card	230	246,323.92
Debit Card	232	232,361.18

Insight: Credit Card brings the most revenue; Debit Card the least.

Q7. Revenue by referral source

```
SELECT ReferralSource,
       COUNT(*) AS orders,
       ROUND(SUM(TotalPrice),2) AS revenue,
       ROUND(AVG(TotalPrice),2) AS avg_order_value
FROM orders
GROUP BY ReferralSource
ORDER BY revenue DESC;
```

ReferralSource	orders	revenue	avg_order_value
Instagram	259	275,285.45	1,062.88
Email	250	261,808.55	1,047.23
Google	241	250,441.48	1,039.18
Facebook	228	250,410.90	1,098.29
Referral	222	226,815.58	1,021.69

Insight: Instagram is the best referral source by both orders (259) and revenue (275,285).

Q8. Yearly revenue trend

```
SELECT STRFTIME('%Y', Date) AS year,
       COUNT(*) AS orders,
       ROUND(SUM(TotalPrice),2) AS revenue
FROM orders
GROUP BY year
ORDER BY year;
```

year	orders	revenue
2023	510	552,643.24
2024	459	480,235.87
2025	231	231,882.85

Insight: Revenue fell from 2023 to 2024; 2025 is a partial year (231 orders), so it is not directly comparable.

Q9. Top 10 highest-value orders (WHERE + ORDER BY)

```
SELECT OrderID, CustomerID, Product, Quantity, TotalPrice
FROM orders
WHERE OrderStatus = 'Delivered'
ORDER BY TotalPrice DESC
LIMIT 10;
```

OrderID	CustomerID	Product	Quantity	TotalPrice
ORD200789	C57276	Tablet	5	3,456.40
ORD200632	C67260	Laptop	5	3,390.80
ORD201065	C47778	Printer	5	3,334.00
ORD200361	C53464	Printer	5	3,299.25
ORD200837	C88029	Chair	5	3,277.75
ORD201079	C89979	Phone	5	3,170.00
ORD200511	C11415	Monitor	5	2,876.20
ORD200578	C50415	Monitor	5	2,830.35
ORD200883	C43595	Printer	5	2,807.40
ORD200769	C13366	Printer	4	2,714.20

Insight: The largest delivered orders are all 4 to 5 units; the top order is a Tablet worth 3,456.40.

Q10. Large orders: TotalPrice >= 2000 and not cancelled

```
SELECT COUNT(*) AS large_orders,
       ROUND(SUM(TotalPrice),2) AS revenue
FROM orders
WHERE TotalPrice >= 2000
  AND OrderStatus <> 'Cancelled';
```

large_orders	revenue
141	358,299.99

Insight: 141 non-cancelled orders of 2,000 or more contribute 358,300 in revenue.

Q11. Revenue by coupon code (NULL = no coupon)

```
SELECT COALESCE(CouponCode,'No Coupon') AS coupon,
       COUNT(*) AS orders,
       ROUND(AVG(TotalPrice),2) AS avg_order_value,
       ROUND(SUM(TotalPrice),2) AS revenue
FROM orders
GROUP BY coupon
ORDER BY orders DESC;
```

coupon	orders	avg_order_value	revenue
FREESHIP	313	1,070.41	335,036.99
No Coupon	309	1,043.37	322,401.41
WINTER15	292	1,035.90	302,483.54
SAVE10	286	1,065.87	304,840.02

Insight: FREESHIP is the most-used coupon, and 309 orders used no coupon at all.

Q12. Products with more than 150 orders (HAVING)

```
SELECT Product, COUNT(*) AS orders
FROM orders
GROUP BY Product
HAVING COUNT(*) > 150
ORDER BY orders DESC;
```

Product	orders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

Insight: Every product has more than 150 orders, so none is filtered out by the HAVING condition.

Q13. Percentage contribution of each product to total revenue

```
SELECT Product,
       ROUND(SUM(TotalPrice),2) AS revenue,
       ROUND(100.0 * SUM(TotalPrice) / (SELECT SUM(TotalPrice) FROM orders), 2) AS pct_of_revenue
FROM orders
GROUP BY Product
ORDER BY pct_of_revenue DESC;
```

Product	revenue	pct_of_revenue
Printer	195,612.61	15.47
Chair	195,620.11	15.47
Laptop	192,126.56	15.19
Tablet	186,568.95	14.75
Monitor	175,651.41	13.89
Desk	167,459.93	13.24
Phone	151,722.39	12.00

Insight: Product revenue is evenly spread: from 12.00% (Phone) to 15.47% (Chair and Printer).

Q14. Cancellation + return rate by product

```
SELECT Product,
       COUNT(*) AS orders,
       SUM(CASE WHEN OrderStatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END) AS cancelled_or_returned,
       ROUND(100.0 * SUM(CASE WHEN OrderStatus IN ('Cancelled','Returned') THEN 1 ELSE 0 END) / COUNT(*), 2
FROM orders
GROUP BY Product
ORDER BY loss_rate_pct DESC;
```

Product	orders	cancelled_or_returned	loss_rate_pct
Monitor	163	71	43.56
Tablet	179	77	43.02
Laptop	173	74	42.77
Chair	178	73	41.01
Printer	181	73	40.33
Phone	156	62	39.74
Desk	170	67	39.41

Insight: Around 40% of orders for every product are cancelled or returned; Monitor is highest at 43.56%.

Q15. Top 10 customers by spend (repeat customers)

```
SELECT CustomerID, COUNT(*) AS orders, ROUND(SUM(TotalPrice),2) AS total_spent
FROM orders
GROUP BY CustomerID
ORDER BY total_spent DESC
LIMIT 10;
```

CustomerID	orders	total_spent
C38840	2	5,723.23
C57276	1	3,456.40
C67260	1	3,390.80
C13877	1	3,384.90
C18404	1	3,370.20
C16775	1	3,353.75
C65986	1	3,352.40
C47778	1	3,334.00
C59183	1	3,322.55
C25276	1	3,313.90

Insight: Almost every customer ordered once; only C38840 ordered twice.

Q16. Realised revenue (Delivered + Shipped only) by product

```
SELECT Product, ROUND(SUM(TotalPrice),2) AS realised_revenue
FROM orders
WHERE OrderStatus IN ('Delivered','Shipped')
GROUP BY Product
ORDER BY realised_revenue DESC;
```

Product	realised_revenue
Printer	82,452.14
Laptop	80,951.66
Chair	71,816.27
Monitor	66,924.44
Tablet	66,622.67
Phone	62,342.05
Desk	57,650.67

Insight: Counting only Delivered and Shipped orders, Printer leads with 82,452 in realised revenue.

Q17. Data quality: NULL coupons, duplicate order IDs, price mismatch

```
SELECT SUM(CASE WHEN CouponCode IS NULL THEN 1 ELSE 0 END) AS null_coupons,
       COUNT(*) - COUNT(DISTINCT OrderID) AS duplicate_order_ids,
       SUM(CASE WHEN ABS(Quantity*UnitPrice - TotalPrice) > 0.01 THEN 1 ELSE 0 END) AS price_mismatch_rows
FROM orders;
```

null_coupons	duplicate_order_ids	price_mismatch_rows
309	0	0

Insight: No duplicate OrderIDs, and TotalPrice always equals Quantity x UnitPrice, so the data is clean. NULL coupons mean no coupon was used.

3. Conclusion

Using SELECT, WHERE, GROUP BY, HAVING, ORDER BY and the COUNT, SUM and AVG aggregates, this analysis turned 1,200 raw order rows into clear answers. The main business concern is the high rate of cancelled and returned orders across all products, while the main strengths are even product demand and strong results from Instagram and FREESHIP-driven orders.